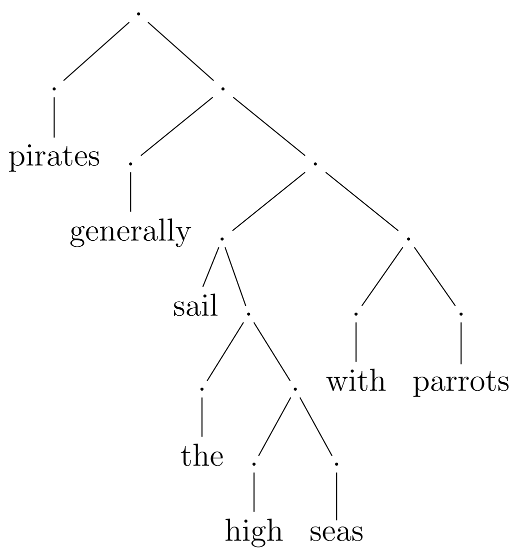
COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 7

6 Formal Models of Language (pjb48)

Consider the following syntax tree:



Context Free Grammar

(*a*) Provide a Context-Free Grammar that could generate the tree. [5 marks]

Answer: Providing a full CFG specification. [1 mark]

$\mathcal{N} = \{S, NP, VP, PP, N, V, P, J, D, A\}$

$\Sigma = \{generally, high, parrots, pirates, sail, the, with\}$

$S = S$

$\mathcal{P} = \{S \rightarrow NP\ VP$

$NP \rightarrow J\ NP \parallel N \parallel D\ NP$ [1 mark]

$VP \rightarrow VP\ PP \parallel V\ NP \parallel A\ VP$ [2 marks]

$PP \rightarrow P\ NP$ [1 mark]

$A \rightarrow generally$

$D \rightarrow the$

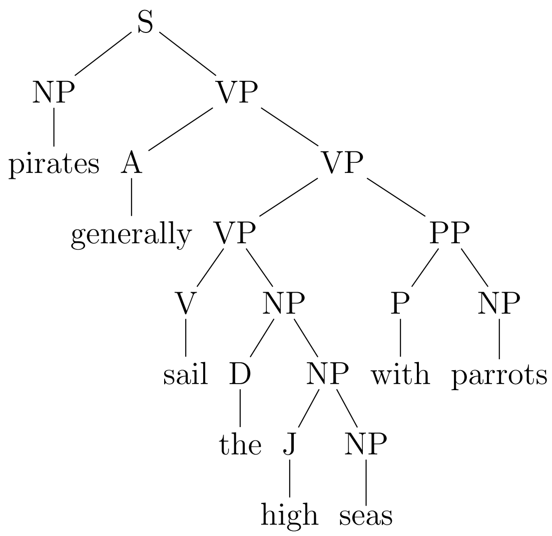
$J \rightarrow high$

$N \rightarrow pirates \parallel seas \parallel parrots$

$P \rightarrow with$

$V \rightarrow sail\}$

Which would fill out the tree as follows:



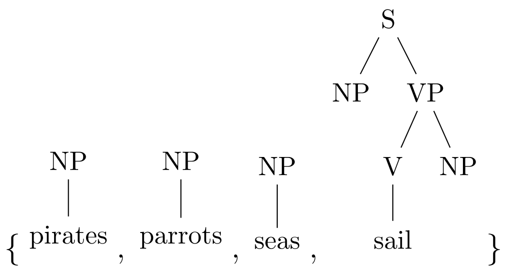
Tree Adjoining Grammar

(*b*) Provide a Tree-Adjoining Grammar that could generate the tree. [5 marks]

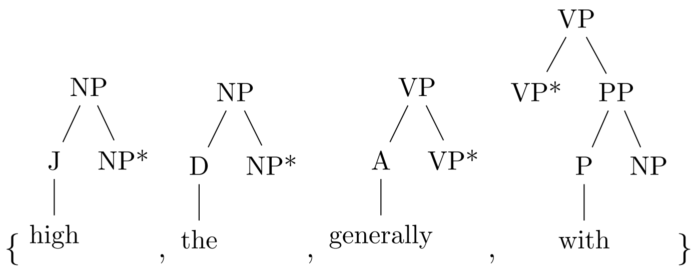
Answer:

$$\mathcal{N} = \{S, NP, VP, PP, N, P, V, A, J, D\}$$
$$\Sigma = \{pirates, generally, sail, the, high, seas, with, parrots\}$$
$$S = \text{S}$$

$\mathcal{I}$ =



$\mathcal{A}$ =

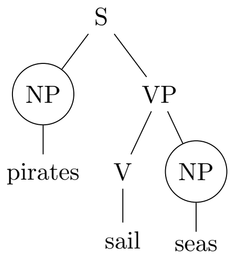


Providing a full TAG specification [1 mark]. The set of initial trees [2 marks].The set of auxiliary trees [2 marks]

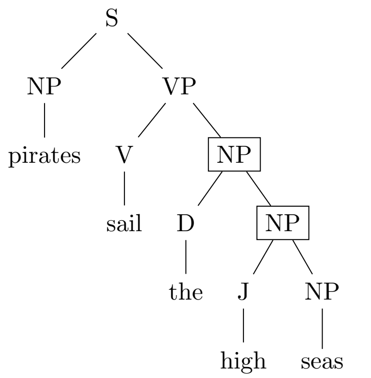
TAG parsing

(*c*) Show how to generate the tree using your Tree-Adjoining Grammar. [5 marks]

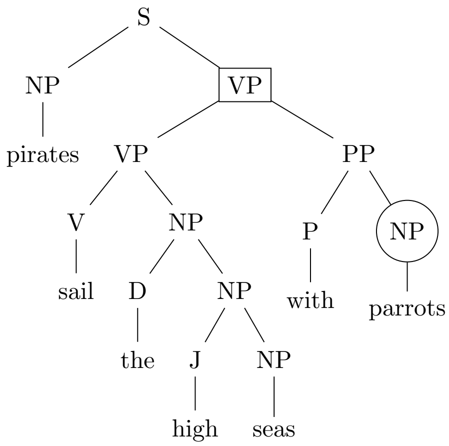
Answer:



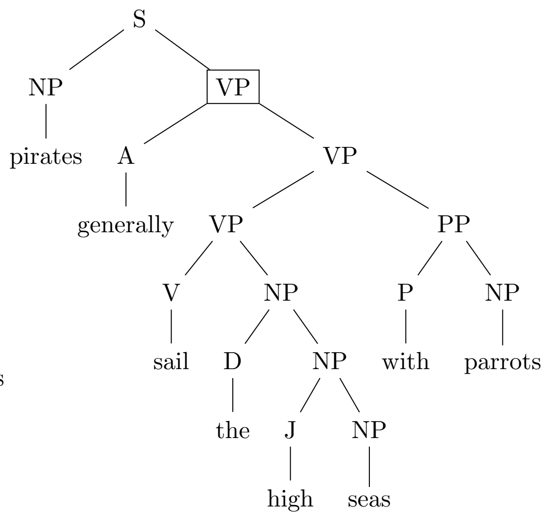
1. two substitutions



2. 2 adjunctions



3. 1 substitution and 1 adjunction

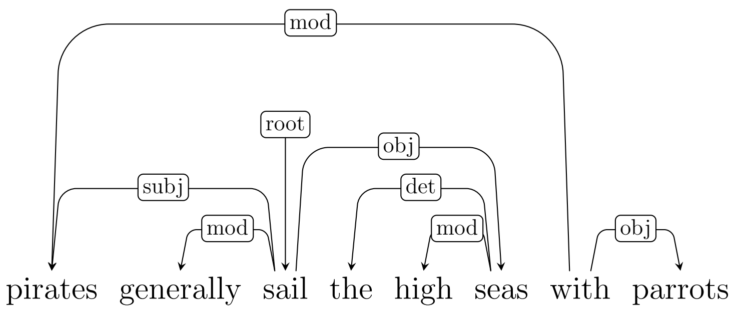


4. 1 adjunction

Answer consistent with your TAG [1 mark]. Knowledge of substitution [1 mark] and adjunction [1 mark]. Correct generation [1 mark]. Some order to generation [1 mark]

GRs

(*d*) Could the following Grammatical Relations be derived from the tree? Explain your answer.



[5 marks]

Answer: Projective grammatical relations weakly equivalent to CFGs [1 mark]. Explanation of the relationship between nodes and dependencies [2 marks]. Notice that parrots is modifying pirates [1 mark] explain that requires another tree [1 mark]
